

ESConv: Predicting Support Strategies and Generating Supporter Replies

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1. The Task

ESConv (Liu et al., 2021) is a corpus of roughly 1,300 emotional-support conversations between a help-seeker and a trained supporter. Every supporter turn carries one of eight support strategies (e.g. Question, Providing Suggestions, Reflection of feelings), and each conversation carries a seeker-reported problem type and emotion. The dataset ships as one JSON file with no predefined split, so the split protocol is itself part of the deliverable.

This report covers two subtasks: **classification** (predicting the supporter’s next strategy from the preceding dialogue, as both an 8-class and a coarser 3-class problem: Exploration / Comforting / Action) and a **generative-AI subtask** (using an LLM to draft the supporter’s response text, conditioned on either the classifier’s predicted strategy or the gold strategy, scored by an LLM judge and by human annotation).

2. Split Protocol

Grouping unit: the conversation. The 1,300 conversations were assigned to train (0.8) / validation (0.1) / test (0.1) at the conversation level, so no conversation contributes turns to more than one side.

Measured gap. To measure what a random-utterance split would leak, I trained a TF-IDF + logistic-regression probe (3-class, class-weighted) under both splits. The reason this model was chosen was its speed, which let me repeat the comparison across 20 seeds, and a low-capacity model is if anything the *more* sensitive leakage probe (Soltaniani & Ghafari).

Split	Macro-F1, seed 42	Macro-F1, 20-seed mean (std)
Random utterance	0.4341	0.4301 (0.0117)
Conversation level (chosen)	0.4217	0.4293 (0.0136)
Gap (random – conversation)	+0.0124	+0.0008

Table 1. Split-protocol gap. Single seed: bootstrap 95% CI $[-0.022, +0.048]$, $p = 0.24$. Twenty paired seeds: 9 positive / 11 negative, sign test $p = 0.75$, Wilcoxon $p = 0.42$.

The leak I expected is not detectable with this probe, possibly a consequence of the context-window design. The conversation split was chosen anyway, since the ESConv authors also split by conversation, which keeps my protocol aligned with theirs (§3.5).

Context window. The input for each supporter turn is the preceding seeker turns plus the supporter’s own earlier turns in the same speaker burst — not the full history. Turns do not alternate cleanly (90.6% of conversations contain a same-speaker run; 52.5% open with the supporter) and conversations run 16 to 120 turns, so a fixed-alternation design breaks and full history grows unboundedly. Speaker tags [seeker] / [supporter] are inserted. Full-history context was not tested; the original paper also bounds context, to five utterances.

3. Classification

3.1 Models

Position-only decision tree: Two features and no text: the turn's relative position in the conversation (0 = start, 1 = end), and its ordinal among the supporter's turns. I built it to measure how far conversational stage alone predicts strategy, before crediting any text model with learning something about the language.

TF-IDF + logistic regression: Unigrams and bigrams, 20k features, class-weighted; a fixed configuration, deliberately not tuned so that evaluating it on test carries no selection risk. It is the cheap, fully interpretable baseline for a transformer.

DistilBERT: BERT — a masked-language-model transformer encoder — distilled to roughly half the size for faster fine-tuning. Fine-tuned end to end with a three-value learning-rate sweep. It was my first transformer; it is kept as a comparison point.

RoBERTa-base: Roughly twice DistilBERT's size, it is a backbone the published ESConv strategy-prediction papers report against. The transformers use a class-weighted cross-entropy loss (sklearn "balanced" weights), due to the skewed label distribution: unweighted, the loss lets the model ignore the smallest classes, and macro-F1, the metric I optimize for, punishes exactly that.

3.2 Results

Task	Model	Test macro-F1
8-class	RoBERTa-base	0.2398
8-class	TF-IDF + LogReg	0.2189
8-class	DistilBERT	0.2174
8-class	Position-only tree	0.1795
3-class	RoBERTa-base	0.4463
3-class	Position-only tree	0.4418
3-class	TF-IDF + LogReg	0.4293
3-class	DistilBERT	0.4269

Table 2. Test macro-F1 on my conversation-level test split (8-class $n = 1,724$; 3-class $n = 1,417$). Accuracy and weighted-F1 for RoBERTa: 0.274 / 0.282 (8-class), 0.451 / 0.458 (3-class).

RoBERTa beats TF-IDF on both tasks, but the two gains are not equally trustworthy. On 8-class the gain (+0.021) is about 1.5× the seed-to-seed noise. On 3-class the gain (+0.017) is about the size of the noise, and a tree that never reads the text comes within 0.005 of it. RoBERTa is still the best number on both tasks, so it is the model I report; but for 3-class under a cost constraint I would ship TF-IDF.

Per-class results:

Label	Precision	Recall	F1	Support
Question	0.492	0.461	0.476	334
Others	0.455	0.391	0.420	307
Providing Suggestions	0.261	0.226	0.242	283
Affirmation and Reassurance	0.235	0.150	0.183	267
Reflection of feelings	0.141	0.218	0.171	142
Restatement or Paraphrasing	0.242	0.119	0.160	126
Self-disclosure	0.130	0.195	0.156	169
Information	0.081	0.167	0.109	96

Table 3. RoBERTa, 8-class, per-class test results.

Label	Precision	Recall	F1	Support
Exploration	0.634	0.457	0.531	602

Label	Precision	Recall	F1	Support
Comforting	0.370	0.468	0.413	436
Action	0.370	0.422	0.395	379

Table 4. RoBERTa, 3-class, per-class test results.

Accuracy (27.4%) and weighted-F1 (0.282) are supported by "Question" and "Others", 37% of the test set. Five of eight classes have F1 below 0.20. My reading is that some strategies are semantically overlapping, and it's hard to separate from *each other*, not just from the distinctive classes. The error analysis (§7) supports this: two of the ten items I read by hand fit two of these labels at once.

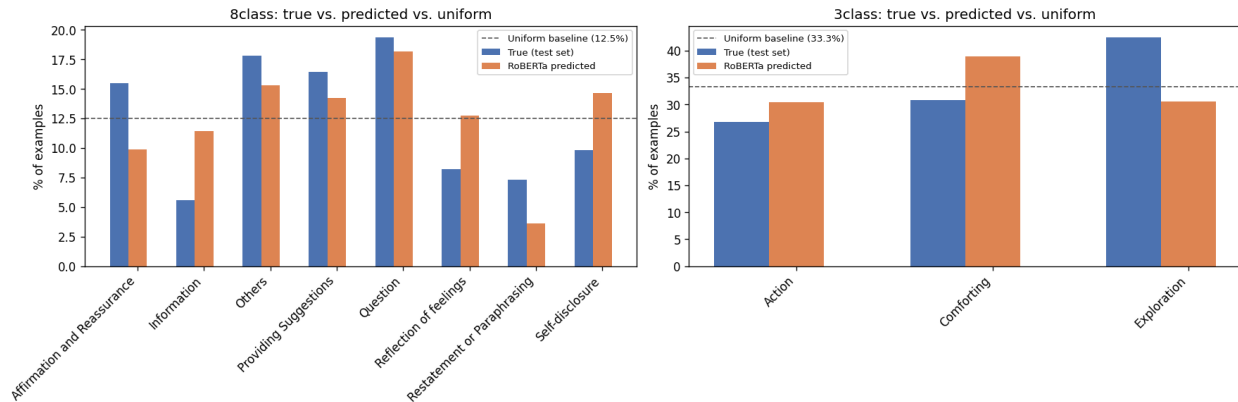


Figure 1. True vs. predicted label share against a uniform baseline, both tasks. Source: 02_classifier_evaluation.ipynb.

The class-weighted loss does what it should: 3-class predictions are pulled toward uniform relative to the true skew (Exploration 42.5% true → 30.6% predicted; Comforting 30.8% → 38.9%).

3.3 The 8 → 3 mapping

Strategy	Mean position	Group
Question	0.289	Exploration
Restatement or Paraphrasing	0.331	Exploration
Reflection of feelings	0.481	Exploration
Self-disclosure	0.510	Comforting
Affirmation and Reassurance	0.539	Comforting
Providing Suggestions	0.592	Action
Information	0.599	Action
Others	0.664	excluded

Table 5. Mean normalized turn position (0 = start, 1 = end) over all 18,376 labeled turns.

There is no one objective way to group the eight strategies into three stages: doing so involves real judgment, and even the original paper's own figure assigns three strategies (Reflection of Feelings, Self-disclosure, and Affirmation and Reassurance) to more than one stage rather than picking a single one for each. The grouping above combines three things: my personal judgment, each strategy's mean normalized turn position, and, that the resulting three groups land in the same order as the paper's own stage figure.

"Others" is excluded rather than force-mapped: it's a heterogeneous catch-all, not a coherent stage. Nearly half its occurrences (45.2%) are lexically openings or closings ("hello," "thanks," "take care") rather than support content, and its position also has by far the widest spread of any strategy (std 0.336 vs. 0.217-0.273).

for every other class) – it isn't concentrated at one point in the conversation the way a real stage would be, but spans both ends, further evidence there's no single stage to assign it to.

3.4 Interpretability / Explainability

The **tree** splits almost entirely on normalized position (importance 0.975 on 8-class, 0.918 on 3-class). With 20 leaves **it never predicts** Information, Reflection, Restatement or Self-disclosure at all — its 8-class score is bought **by giving up** on four classes, not by learning them. On 3-class, where the groups were built around position, it needs only 7 leaves to come within 0.005 of RoBERTa.

The **linear model's** coefficients are its explanation. Two things stand out. The top words for each 3-class group track whichever 8-class component has more training data (Exploration's top words are Question's; Action's are Providing Suggestions'), not an even blend. And many top words are position markers — *hi, hello, how are* for Question; *bye, thanks, take care* for Others — rather than content. Three independent methods (tree, coefficients, the opening/closing phrase counts in the exploration notebook) point at the same thing: conversational stage carries a large share of the signal.

RoBERTa was probed with Integrated Gradients, with a token-occlusion cross-check.

RoBERTa interpretability, 8-class: Integrated Gradients + occlusion cross-check

■ pushes toward the predicted class ■ pushes away intensity = magnitude

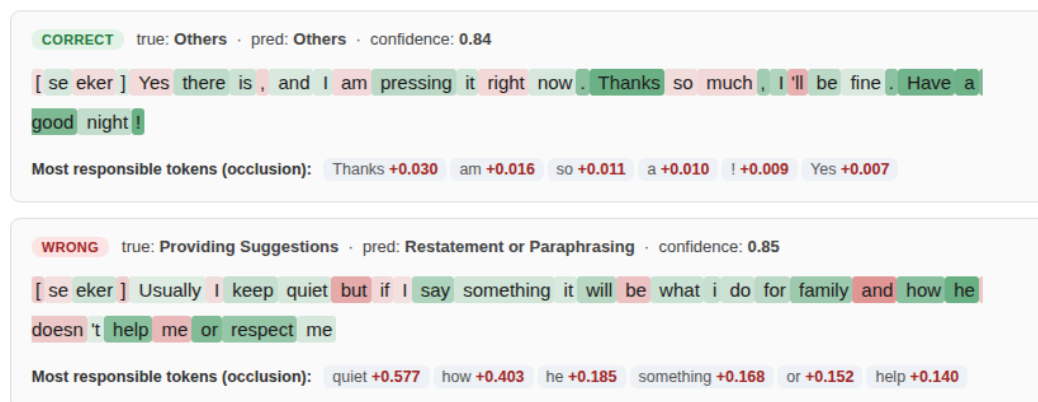


Figure 2. Integrated Gradients and occlusion, 8-class RoBERTa: a confident-correct prediction (idx 1351) and a confident-wrong one (idx 987). Source: 02_classifier_evaluation.ipynb.

On the wrong prediction (a suggestion mislabeled as a paraphrase) the two methods disagree: occlusion blames *quiet* (removing it drops confidence by 0.58); IG gives *quiet* 9% of its peak attribution and points at *he*. When two attribution methods disagree on the same error, neither top token is "the" reason. On the correct prediction no token dominates — the largest occlusion drop is 19× smaller — and the top tokens are closing phrases and punctuation: an easy turn decided by many weak cues. That is also a warning. The confident-correct pool is dominated by closings, so RoBERTa is most confident exactly where the label carries least information.

3.5 Calibration against published numbers

The original paper does not report a standalone strategy classifier; strategy prediction happens inside its generative "Joint" model. EmoDynamix (Wan et al., 2025) does report 8-class macro-F1:

Model	Macro-F1
EmoDynamix	27.70
TransESC	26.28
LLaMA3-8B	25.91
BART (fine-tuned)	25.66
RoBERTa (fine-tuned)	25.04
RoBERTa-base (This report)	23.98
DistilBERT (This report)	23.12
ChatGPT (zero/few-shot)	18.14

Table 6. 8-class strategy prediction macro-F1. Their rows: the ESConv release split and a five-utterance window. My rows: my own conversation-level split and the burst window. Comparable in scale, not like-for-like.

My RoBERTa lands one point under theirs on the same backbone with a much simpler recipe; I read that as the pipeline being in the right place, not as a ranking. The useful information in this table is the ceiling: purpose-built research systems get to 27–28, so the 8-class task is hard for everyone.

4. Generative Subtask

4.1 Design

For 200 test-set supporter turns — the 32 hand-scored items plus 168 sampled proportionally by gold strategy. The generator is Claude Haiku 4.5; the judge is Gemini 2.5 Flash. I did not use the full 1,724-turn test set for cost reasons.

4.2 Prompt designs

Four prompt levels, each adding scaffolding: **L1** a bare "You are a chat assistant" with the strategy named — a naive floor; **L2** adds a one-line strategy definition and a supporter persona; **L3** adds do/don't guidance, a banned-phrase list, and one train-set example per strategy; **L4** adds a second, deliberately different example and an instruction not to copy either.

To pick two levels I ran all four on a 32-item set stratified by gold strategy × whether RoBERTa predicted it correctly (16 cells × 2), scored the 128 responses myself blind to Gemini's scores, and compared. Both raters rank the levels $2 > 4 \approx 3 > 1$. I carried **L2** and **L4** to the full run.

4.3 The judge

The judge scores each response on quality and strategy adherence (1–5 each). Each response is scored alone, never side by side, and all responses share the 40-word cap — which removes the position and verbosity biases named in Zheng et al. (2023). Following the same paper, the JSON output places a short "reasoning" field before the two scores, with an explicit instruction not to pick the numbers first and justify them afterward.

Dimension	Cohen's κ	Quadratic-weighted κ
Quality	0.013	0.063
Strategy adherence	0.033	0.053

Table 7. My scores vs. Gemini's, 128 responses (32 items × 4 levels). Source: full32_human_scores.csv, full32_judge_scores_gemini.json.

Both dimensions sit at chance-level agreement. Gemini's mean is above mine at every level, with the larger, more consistent gap on quality (+1.0 to +1.5) and adherence close to matched, even flipping in my favor at level 1 (Table 8). That points to leniency — a systematic upward offset — rather than disagreement about which responses are good.

Level	My quality	Gemini quality	My adherence	Gemini adherence
1	3.28	4.78	4.34	4.19
2	3.97	5.00	4.72	4.97
3	3.78	4.84	4.66	4.72
4	3.84	4.81	4.59	4.88

Table 8. Mean scores by level, same 128 responses.

κ scores each response individually while the ranking is an aggregate, so per-response noise cancels there: we agree on order ($2 > 4 \approx 3 > 1$). One pattern the blended mean hides: I rate adherence above quality at every level — the responses mostly execute the right strategy without always being good chat messages — where Gemini shows no such gap.

My scores are also lower for items where RoBERTa's predicted strategy did not match the gold label, by +0.44 on average across the four levels: a wrong strategy signal measurably hurts the reply. Gemini shows no such effect; its sign flips between levels.

The stricter rubric this motivated — pushing the full 1–5 range instead of clustering near the top, and a hard penalty for leaked system or formatting artifacts — was carried into the 200-item run (§4.4).

4.4 Final Results

genai_final_level2.xlsx and genai_final_level4.xlsx, 200 rows each, one row per turn. The 32 pilot items reused their L2 and L4 generations; the 168 new items were generated fresh; all 400 responses were judged under the stricter rubric. Responses were left as generated, with a word_count column so any overrun of the 40-word cap is visible rather than trimmed.

Level	Quality	Adherence	n
L2	4.775	4.515	200
L4	4.880	4.795	200

Table 9. Gemini scores, final run (stricter rubric; not comparable score-for-score to Table 8).

L4 is higher on both dimensions under the stricter rubric, while the human pilot preferred L2. With $\kappa \approx 0$ the judge alone cannot settle that, and I leave it open.

Level	Matched mean	n	Mismatched mean	n	Gap
L2	4.746	65	4.596	135	+0.150
L4	4.931	65	4.793	135	+0.138

Table 9b. Same 400 responses, split by whether RoBERTa's predicted strategy matched the gold strategy. Mean of quality and adherence.

A small gap in the same direction survives at both levels: replies generated under a wrong strategy score a little lower. I read it as suggestive rather than clean. The mismatched items are also the ones the classifier found hard, and §7 shows those skew toward closings and thin content, so item difficulty is confounded with the strategy signal — the same caveat as in §4.3.

5. Failed Experiments

Learning-rate sweep for DistilBERT. Three values (5e-5, 3e-5, 2e-5), both tasks. The band was narrow (8-class 0.2194–0.2312; 3-class 0.4261–0.4348), and every run showed the same curve: validation F1 peaks by epoch 1–4 while training loss keeps falling. What it taught me: the binding constraint was capacity against ~14k examples, not the learning rate.

Regularization grid to fix that overfitting. Weight decay {0.01, 0.05, 0.1} × frozen layers {0, 2} at the winning LR. None of six beat the original (best 0.2267 vs. 0.2312). Diagnosis: I changed the LR schedule at the same time (linear decay → constant with warm-up), and the one grid cell identical to the original except for the schedule scored 0.2190 — so the schedule alone cost 0.012 and the linear decay had been doing implicit regularization. Two changes at once left me unable to attribute the result to either; the clean version varies one thing.

Cost-complexity pruning for the tree. Selecting `ccp_alpha` by validation F1 never converged: validation kept improving down to a 3,390-leaf tree (~4 examples per leaf). With two features, one continuous, finer splits always find a little more signal. I replaced it with an explicit leaf cap and minimum leaf size, at a cost of 0.011 on 8-class test.

6. Where an AI Tool Gave Me Something Wrong

- **No speaker separation in the model input.** The context pipeline was built and presented as complete with no marker between seeker and supporter turns, though an unused tagging option existed. I caught it inspecting a raw training example. A TF-IDF ablation confirmed it mattered (+0.003 macro-F1 on 8-class, +0.013 on 3-class), and both RoBERTa checkpoints were retrained with tags; on 3-class the fix moved test macro-F1 from 0.4395 to 0.4463 and flipped the tree-vs-RoBERTa ranking.
- **A turn-index bug.** Turn position was found by value equality (`list.index`), which returns the *first* identical turn. 97 conversations (7.5%) contain duplicate turns; 100 labeled turns got the wrong position — the feature behind 92–98% of the tree's decisions. Found only when I asked for a self-audit; fixed with identity-based lookup, then verified by an invariant test over all 1,300 conversations that also catches cross-conversation context leakage (0 violations).
- **sklearn's tokenizer default.** `\w\w+` treats the apostrophe as a boundary: *that's* → *that* + a dropped *s*, so *re* and *s* showed up as "important" words. Not a project bug, but unflagged. Fixed at all five call sites; no ranking changed.
- **Two LLM-API failures caught only by reading output.** Gemini 2.5 Flash spent its 250-token output budget on hidden thinking, truncating 128 of 128 judge replies to a few words; fixed with `thinking_budget=0`. Separately, the generator appended its own word count ("*(14 words)*") and, at L1 only, a markdown header — stripped post hoc rather than by adding instructions to L1, which would have blurred what L1 tests.
- **A generation design that measured nothing for a third of the items.** My first run generated each of 100 turns under both the predicted and the gold strategy, 400 generations. For 30 of the 100 turns the prediction was correct, so the two "conditions" were the same prompt sent twice: no contrast, just duplicated rows. I caught it while auditing the `xlsx`. I rebuilt the run as 200 turns under the predicted

strategy only and moved the predicted-vs-gold comparison to the matched/mismatched split (Section 4.4), where every item contributes one observation.

7. Error Analysis: Ten Items by Hand

I started with the ten highest-confidence wrong predictions and noticed they were nearly all closings. Checking against all 1,251 wrong test predictions confirmed the bias: the model's *Others* prediction accounts for 66.7% of the top 1% most-confident errors against an 11.5% base rate. I reselected — the highest-confidence error per distinct (gold, predicted) pair — so the ten cover different confusions.

idx	Supporter turn (gold → predicted, conf.)	My read
987	"I'm sorry about what you're going through. It might be a good idea to take some time away from him... Do you have friends or family near you?" (Providing Suggestions → Restatement, 0.85)	Clean model error. Good reply, correct label; the model keyed on the opening reflective sentence.
1624	"So you lost your job that is why you are sad right?" (Restatement → Question, 0.88)	Both labels fit. A better supporter would have unpacked the statement rather than echoing it as a question.
648	"Oh no! I'm sorry to hear that. Would you be willing to talk about it with me?" (Reflection → Question, 0.79)	Gold fits, but I would have asked a clarifying question here myself; overlap, not error.
231	"Absolutely." — answering "Is this something you've done to make money?" (Restatement → Self-disclosure, 0.88)	Answering a personal question about yourself is self-disclosure. I side with the model.
1317	"you too!" — final turn (Providing Suggestions → Others, 0.86)	Checked the raw JSON for an indexing bug: the label is as recorded. Annotation noise.
789	"You're welcome. You be safe as well." (Reflection → Others, 0.83)	Reciprocal closing; gold credits strategy content that isn't there.
241	"You too." (Restatement → Others, 0.82)	No mechanism I can find for the gold label. Unexplained noise.
924	"Have a good time with your friends." (Affirmation → Others, 0.81)	More content than "you too", but still a closing remark.
285	"(You need to finish from your end .. I can't :))" (Information → Others, 0.84)	Chat-tool session mechanics leaking into the transcript. Not support content.
1226	"you too! Can you press quit first cos I can't from this end" (Self-disclosure → Others, 0.82)	Same platform artifact as 285.

Table 10. The ten hand-read items (8-class RoBERTa). Reading the supporter's actual reply, not only the model's input, they split four ways:

- **One clean model error** (987): the model read the first clause and stopped. This is also the item where IG and occlusion disagreed (Figure 2).
- **Two genuine overlaps** (1624, 648): the single-label frame forces a choice the text does not make; a multi-label formulation is the obvious follow-up.
- **Five annotation-noise cases** (231, 1317, 789, 241, 924): bare closings labeled with substantive strategies. The model's *Others* is the more defensible read.
- **Two platform artifacts** (285, 1226): interface instructions, not support content.

Seven of ten "errors" are about the labels, not the model, which reframes Table 3: part of the floor on the small classes is label quality, which no classifier recovers. The one clean error also exposed a gap in the

generation prompts — none asks the model to restate the seeker's message before replying; that step exists only in the judge prompt. An "understand, then respond" level is the first thing I would test next.

8. Cost and Latency

Stage	Model	Calls	Input tok	Output tok	Cost	Wall-clock	s/call
Generation, 168 new turns	Claude Haiku 4.5	336	89,900	13,572	\$0.16	389 s	1.16
Generation, 32 pilot turns (reused)	Claude Haiku 4.5	64	17,272	2,609	\$0.03*	82 s	1.28
Judging, all 400 (strict rubric)	Gemini 2.5 Flash	400	279,798	30,151	\$0.16	421 s	1.05
Total		800			\$0.35	14.9 min	

Table 11. Instrumented per-call figures for the responses in the deliverable files. *Paid during the pilot, not this run. Pricing at time of run: Haiku \$1 / \$5 per M tokens in / out; Gemini 2.5 Flash \$0.30 / \$2.50. The judge's input tokens roughly doubled per call versus the pilot (355 → 700) because of the longer rubric.

Per response (generation + judging): about \$0.0009 and 2.2 s. The full 1,724-turn test set at two levels (3,448 responses) would have cost about \$3 and 2.1 hours sequentially.

At 100× this run (20,000 turns, 80,000 calls) the money is still small (~\$35); the problem is time and trust.

Time: the pipeline is sequential and dominated by network round-trip, so the first change is concurrency and provider batch APIs. The second is skipping calls: opening and closing turns are lexically predictable (§ 3.4, 7) and generating a "you too" costs the same as generating a suggestion. Trust: at 40,000 judge calls the blind human scoring that exposed the ceiling effect here does not scale, so I would replace it with a stratified spot-check — a fixed few hundred items re-scored per batch, with κ tracked as a drift alarm rather than measured once.

References

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